

Stage 1 Dynamic Comparison: IEEE 39 Base Case vs. mix60 IBR Case

Spectral IBR validation note

Generated from ANDES outputs in `outputs/stage1_td_compare/mix60`

Abstract

This note documents the Stage 1 time-domain comparison between the original IEEE 39 dynamic case and the converted `mix60` inverter-based resource (IBR) case. The objective is not to claim that the IBR case is fully validated, but to check whether ANDES is integrating the intended IBR dynamic models and whether the response to a common disturbance is physically interpretable. The final scenario disconnects load `PQ_7` at bus 16 at $t = 12$ s. Both cases run to 20 s. The response has the expected sign for a load rejection: frequency and voltages increase. However, the `mix60` case shows large voltage excursions, including buses above 1.10p.u., so the result should be treated as a successful simulation run with engineering warnings, not as a final proof of a healthy IBR benchmark.

1 Purpose of the Check

The question being tested is:

Does the converted `mix60` case actually simulate IBR dynamics in ANDES, and do the plotted signals make electrical sense for the selected event?

The answer is conditional. ANDES is using the intended IBR models, and the qualitative direction of the response is correct. The case is not yet clean enough to call “perfect”, because the converted IBR case has stronger voltage transients than the all-synchronous base case.

2 What Changed in the Scenario

The source workbooks are:

- Base case: `data/raw/ieee39_full.xlsx`.
- IBR case: `data/processed/mix60/mix60.xlsx`.

The scenario workbooks generated for this comparison are:

- `outputs/stage1_td_compare/mix60/base_toggle_pq7.xlsx`.
- `outputs/stage1_td_compare/mix60/mix60_toggle_pq7.xlsx`.

The original workbook includes a disabled `Toggler` for disconnecting `GENROU_1`. That event was not used for this validation because it is a large generator outage and caused early termination in the IBR case. A failed large-contingency run is useful information, but it is not a clean first check for whether the IBR conversion is wired correctly.

The final scenario keeps the generator toggler disabled and adds one active load toggler:

Field	Value
Model	PQ
Device	PQ_7
Bus	16
Active power	3.29 p.u.
Reactive power	0.323 p.u.
Event time	12.0 s
Action	Disconnect load

If the system base is 100 MVA, this is approximately a 329 MW and 32.3 MVar load rejection. This is not a tiny perturbation; it is a meaningful disturbance.

3 Experiment Diagram

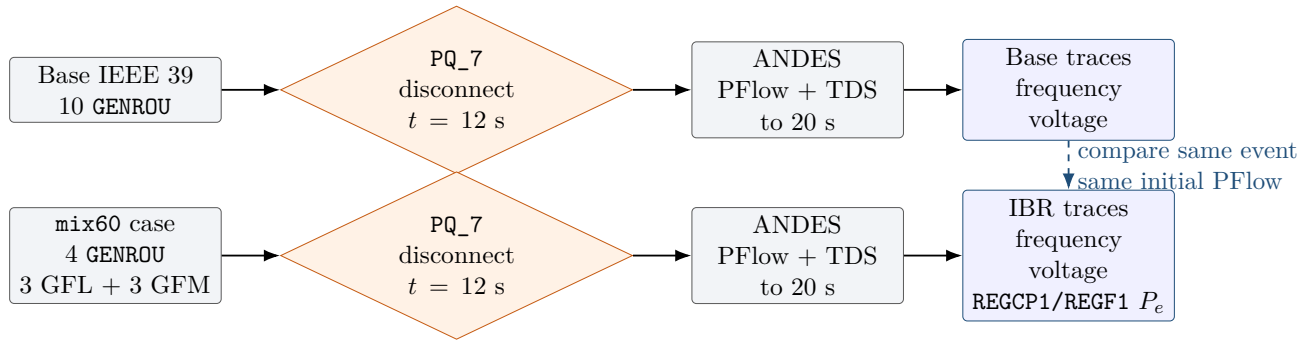


Figure 1: Stage 1 dynamic comparison workflow. The disturbance is identical in both workbooks; only the dynamic resource mix changes.

4 IBR Models Actually Present in ANDES

The mix60 case is not only a static spreadsheet change. ANDES loads dynamic IBR models:

Model class	Base case count	mix60 count
GENROU	10	4
REGCP1 grid-following IBR	0	3
REGF1 grid-forming IBR	0	3
REECA1 electrical controller	0	3
REPCA1 plant controller	0	3
PLL1 phase-locked loop	0	3

The converted generators are:

- GFL: generators 5, 7, and 9 at buses 34, 36, and 38.
- GFM: generators 4, 6, and 8 at buses 33, 35, and 37.
- Synchronous machines retained: buses 30, 31, 32, and 39.

5 Main Numerical Results

Metric	Base case	<code>mix60</code> IBR case
Final simulation time	20.0 s	20.0 s
Max frequency delta, selected buses	0.00131 p.u.	0.00179 p.u.
Approx. max frequency delta	0.079 Hz	0.107 Hz
Bus 16 max voltage delta	0.0199 p.u.	0.0715 p.u.
Bus 35 max voltage delta	0.0150 p.u.	0.0970 p.u.
Max REGCP1 P_e delta	–	0.3752 p.u.
Max REGF1 P_e delta	–	0.6756 p.u.
Buses exceeding 1.10 p.u. during TDS	0	20

6 Interpretation of the Plots

6.1 Frequency

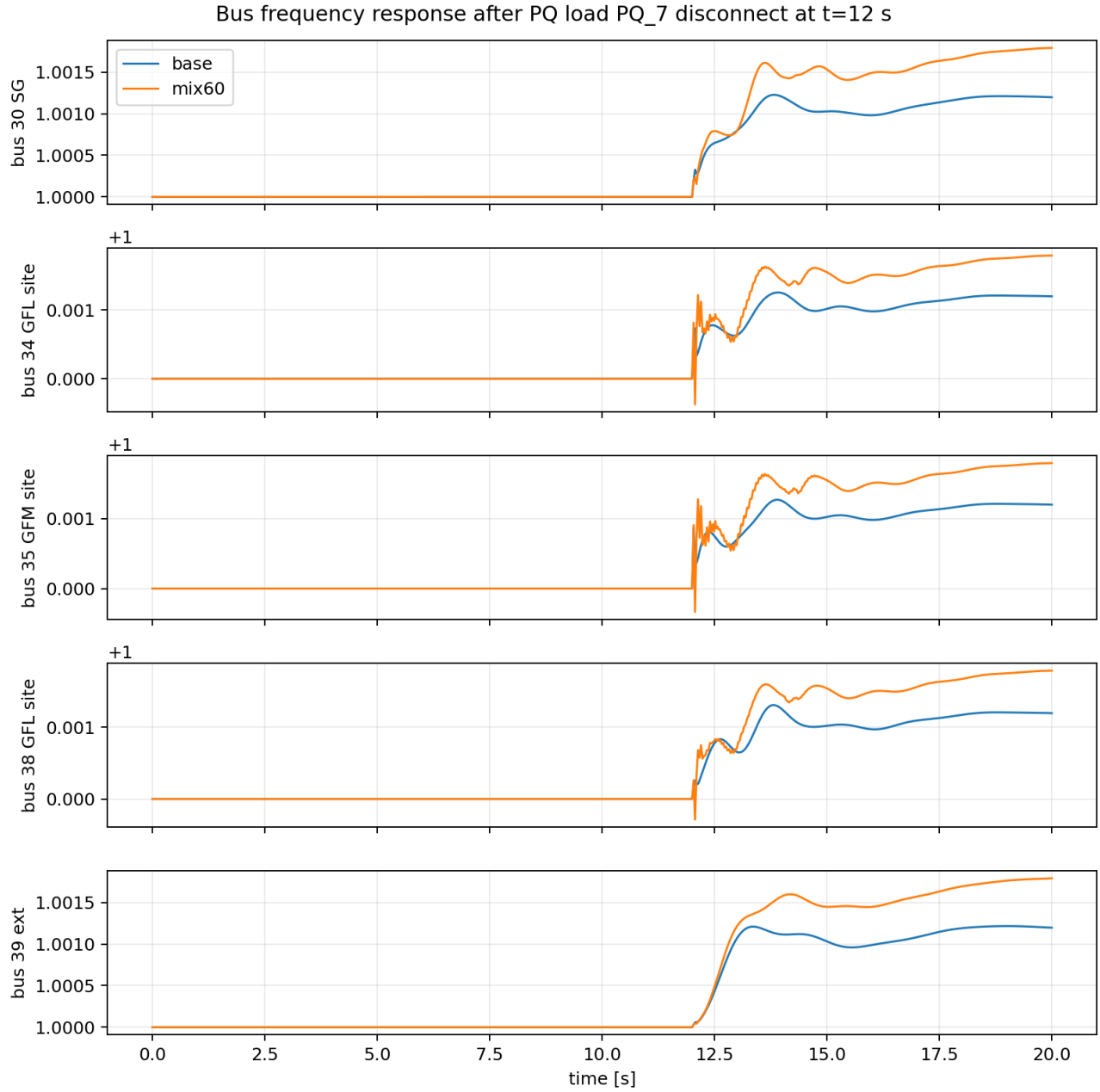


Figure 2: Bus frequency response after disconnecting PQ_7. The positive frequency change is expected because the event removes load. The `mix60` case has a larger final frequency deviation than the base case.

The sign of the frequency response is correct. Removing load causes mechanical input to exceed electrical demand until controls rebalance the system, so frequency rises. The `mix60` case rises more than the all-synchronous base case, which is consistent with lower synchronous inertia and different active-power control behavior.

6.2 Voltage

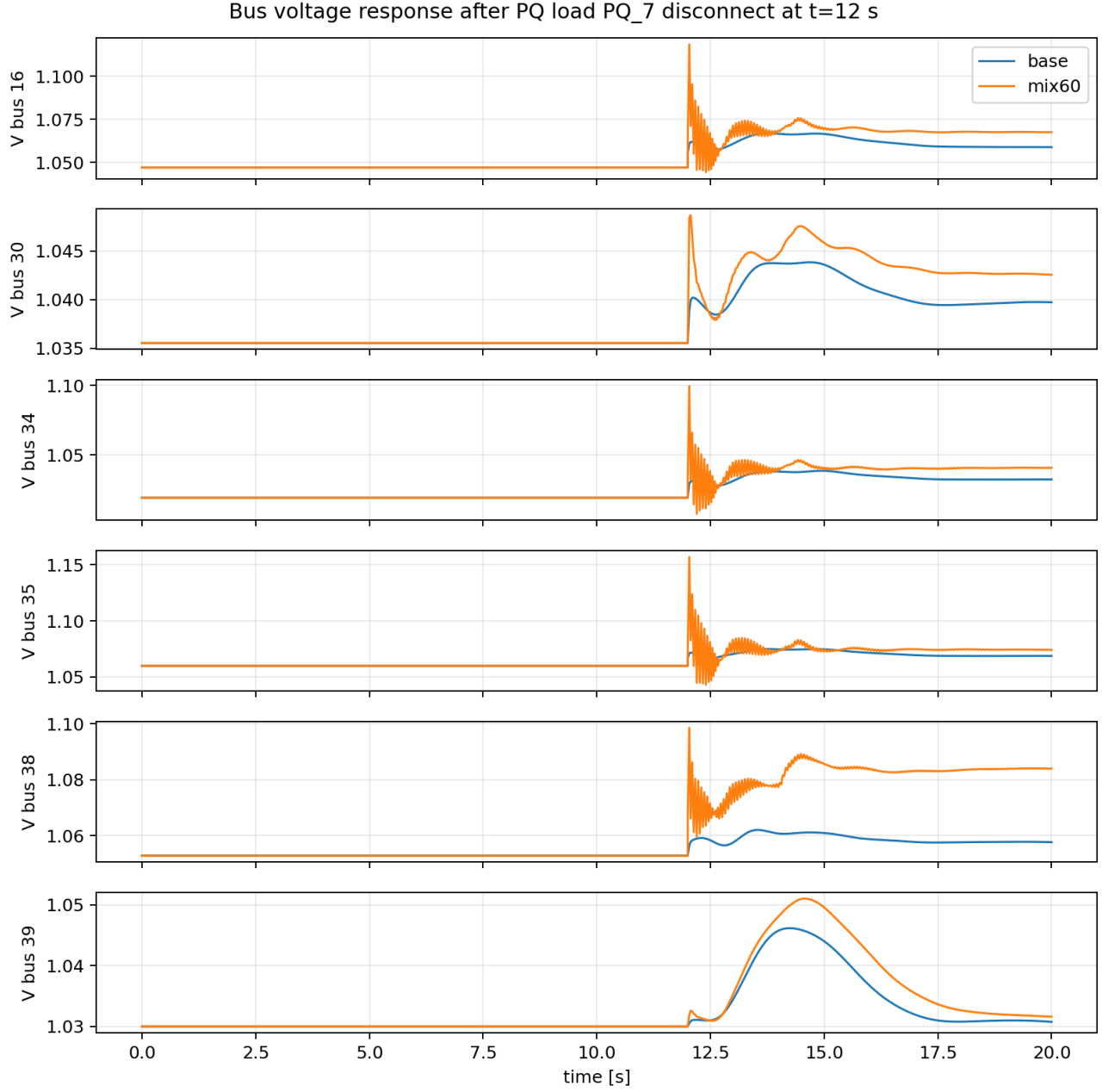


Figure 3: Voltage response at the disturbance bus and selected generator/IBR buses. The `mix60` case has materially larger voltage excursions.

The voltage response also has the expected sign. A load rejection reduces local current and reactive demand, so voltages tend to rise. The concern is the magnitude. In the `mix60` run, bus 35 reaches about 1.157p.u., and 20 buses exceed 1.10p.u.. This is not a numerical crash, but it is an engineering warning.

6.3 IBR Active Power

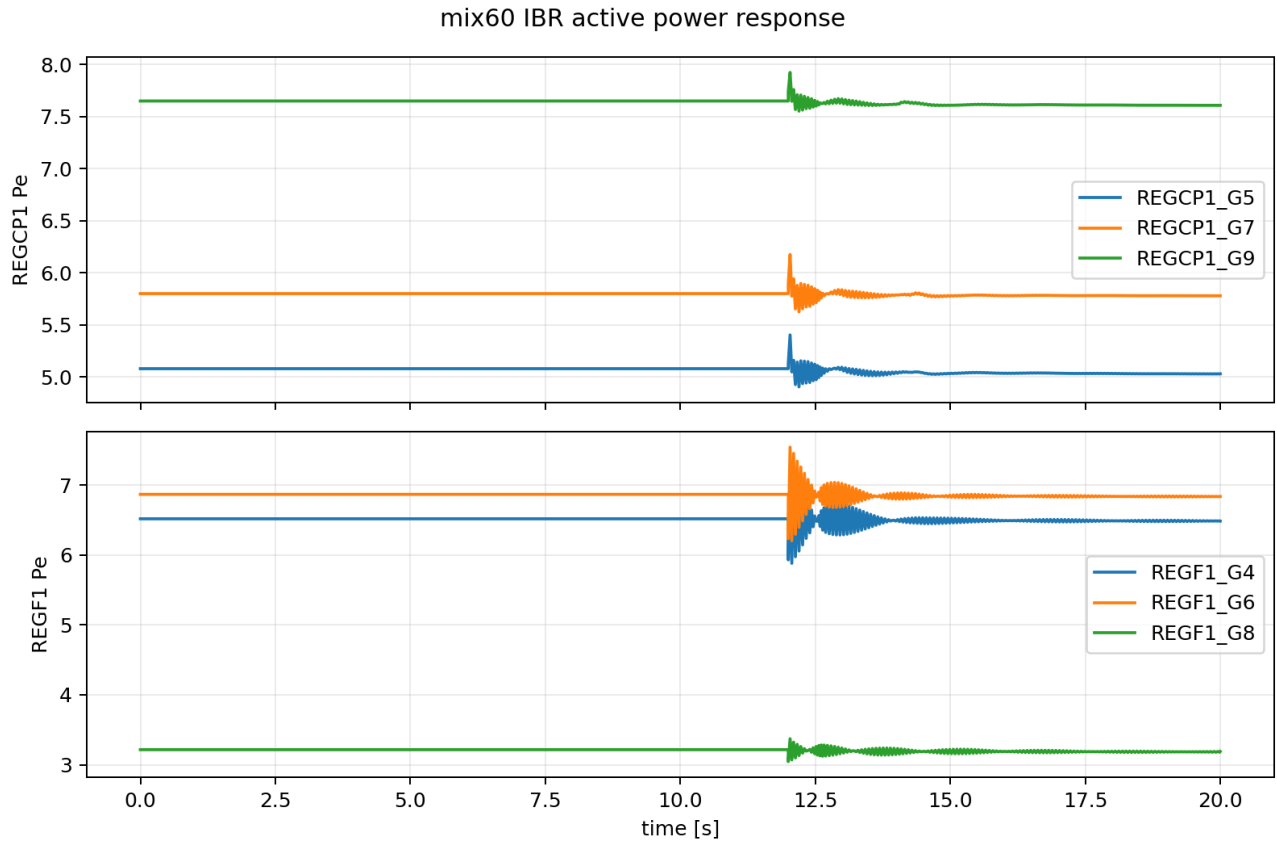


Figure 4: IBR active power response. Both GFL and GFM devices respond after the event; oscillations are visible and then damp out.

The REGCP1 and REGF1 traces confirm that the IBR devices are participating dynamically. The active-power signals move immediately after the event and then settle. This supports the conclusion that ANDES is integrating the IBR controls rather than merely preserving a static power-flow injection.

7 Engineering Concerns

1. The selected event is a load rejection of roughly 329 MW, not a small signal test. It is useful, but should not be the only validation scenario.
2. The mix60 voltage response is high. A production validation gate should flag or fail cases with voltage maxima above chosen thresholds, for example 1.10p.u. warning and 1.15p.u. failure.
3. Some IBR controller limits are very broad, such as high current and reactive-power limits in parts of the GFL control chain. That can make the simulation mathematically robust while still being physically optimistic.
4. The severe original event, disconnecting GENROU_1, caused early termination in prior runs. That result should be documented separately as a contingency stress-test failure, not hidden.

5. A previous attempt to use an **Alter** load step on **Ppf** and **Qpf** did not excite the default load model, because default PQ loads are converted to impedance/current/power portions for TDS. It is safer for this check to use **Toggler** on a real PQ load device.

8 Conclusion

The current Stage 1 comparison demonstrates that:

- The base and **mix60** cases start from the same solved power-flow condition.
- The **mix60** case is using dynamic IBR models in ANDES.
- The response direction to **PQ_7** disconnection is physically interpretable: frequency and voltage rise.
- The IBR active-power traces move and damp after the event.

The result should be reported as:

The **mix60** IBR case simulates successfully in ANDES for the **PQ_7** load-disconnection scenario, and the qualitative response is consistent with load rejection. However, the voltage excursions are high and must be treated as engineering warnings. The case is suitable for continued method development, but not yet as a fully validated benchmark without additional voltage-limit, controller-limit, and smaller-disturbance tests.